Shubham Sah

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Education

Indian Institute of Information Technology (IIIT) **Bhopal, India**
Bachelor of Technology in Computer Science and Engineering (CGPA: 7.48) *Aug. 2023 – May 2027*

Experience

AlgoUniversity **Hyderabad, India**
MTS Intern (Backend & Systems Engineering) *May 2026 – Present*

- **Pioneer** the development and scaling of backend microservices, increasing throughput and system reliability under load.
- **Automate** operational pipelines using Python and Bash scripting, reducing manual workflows by 40%.
- **Diagnose** system bottleneck telemetry and logs, optimizing microservice latency and maximizing service uptime.

Indian Institute of Information Technology (IIIT) **Bhopal, India**
Undergraduate Researcher (Quantum-Safe ZKP) *Aug 2025 – May 2026*

- **Designed** a lattice-based Zero-Knowledge Proof (ZKP) protocol in Python, reducing verification latency by 40% for batch processing.
- **Orchestrated** a hybrid blockchain storage model utilizing off-chain vectors and on-chain Merkle roots, improving data retrieval speeds by 25%.

Maulana Azad National Institute of Technology (MANIT) **Bhopal, India**
Research Intern *Nov. 2023 – Apr. 2024*

- **Deployed** C++ testing frameworks to validate algorithmic efficiency, achieving 90%+ accuracy across core pathfinding services.
- **Streamlined** system telemetry analysis to pinpoint operational bottlenecks, decreasing false positive detections by 35% and boosting service uptime by 15%.

Projects

Remote Browser Control System | ***React, Node.js, Docker, WebSockets, Puppeteer*** GitHub | Live

- **Architected** a remote browser control pipeline using Node.js, WebSockets, and Puppeteer/CDP to stream high-performance Chromium screencasts onto a React canvas.
- **Integrated** real-time mouse coordinate normalization and keyboard key emulation on an HTML5 canvas, reducing user interaction latency below 100ms.
- **Resolved** connection socket leaks and zombie container processes by enforcing strict WebSocket-to-lifespan container bindings.

Carbon Ingestion & Review Ledger | ***Django, React, SQLite, REST APIs, Render*** GitHub | Live

- **Engineered** a multi-tenant carbon data ingestion engine supporting 10,000+ records, applying daily billing proration and airport Haversine distance calculations.
- **Constructed** a self-healing Ingestion Sandbox UI and Facility Wizard, reducing manual CSV/JSON data correction errors by 60%.
- **Implemented** an immutable chronological audit trail with visual before/after diffs, verified by a robust suite of unit tests.

Skills & Expertise

Infrastructure & DevOps: Cloud Computing (AWS/GCP/Render), Docker, Linux/Bash Scripting, System Architecture
Security & Backend: Python, Django, C/C++, RESTful APIs, Applied Cryptography, Zero-Knowledge Proofs
Data & Databases: SQL (SQLite/MySQL), MongoDB, Data Analytics, Data-driven Decision Making
Developer Tools: Git/GitHub, VS Code, Postman, Agile Methodologies, Cross-functional Collaboration, LaTeX
Core Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems

Publications

- **Quantum-Safe ZKP Architecture for Secure Verification** — Authored a research paper detailing a lattice-based Zero-Knowledge Proof protocol; submitted to ***Cluster Computing Journal***.
- **Arduino-Based Robotic Car** — Published an IEEE paper on real-time embedded hardware control and ultrasonic sensor obstacle avoidance; ***IEEE SCEECS 2024***.

Achievements

- **Solved** 670+ algorithmic challenges across LeetCode, Codeforces (Rating 1863), and CodeChef (5-Star) to optimize complex data structures.
- **Committed** open-source code enhancements to distributed developer repositories during GitScript Summer of Code (GSSoC).
- **Secured** top 15 rank out of 500+ students in competitive coding at IIIT Bhopal.